ANSWER KEY: WEEKS 7, 8, 9
H524 Introduction to Biostatistics - Fall 2025
INSTRUCTOR USE ONLY

Week 7: Nonparametric Methods (5 questions, 5 points)

Question 1: When to Use Nonparametric Tests (1 point)

Correct Answer: B
The data do not meet the assumptions required for parametric tests (e.g., normality)

Question 2: Sign Test vs. t-test (1 point)

Correct Answer: B
Sign test
Explanation: Pain score differences are highly skewed, violating normality assumption for paired t-test.

Question 3: Mann-Whitney U Test (1 point)

Correct Answer: C
Independent samples t-test (two-sample t-test)

Question 4: Kruskal-Wallis Test (1 point)

Correct Answer: B
Kruskal-Wallis test
Explanation: Right-skewed data violates normality; Kruskal-Wallis is nonparametric alternative to one-way ANOVA.

Question 5: Ranking Data (1 point)

Correct Answer: B
Tied values receive the average of the ranks they would have occupied

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Week 8: Contingency Tables / Chi-Square (9 questions, 9 points)

Question 1: Contingency Table Basics (1 point)

Correct Answer: C
Two categorical variables

Question 2: Expected Frequencies (1 point)

Correct Answer: A
(Row total × Column total) / Grand total

Question 3: Chi-Square Test Assumptions (1 point)

Correct Answer: B
All expected cell frequencies should be at least 5

Question 4: Degrees of Freedom (1 point)

Correct Answer: B
6
Calculation: df = (r-1)(c-1) = (3-1)(4-1) = $2 \times 3 = 6$

Question 5: Interpreting Chi-Square Results (1 point)

Correct Answer: B
There is a significant association between the variables
Explanation: $p = 0.032 < 0.05$, so we reject the null hypothesis of no association.

Question 6: Relative Risk (1 point)

Correct Answer: B
The ratio of the risk of disease in exposed vs. unexposed groups

Question 7: Odds Ratio (1 point)

Correct Answer: B
The odds of disease in the exposed group are 2.5 times the odds in the unexposed group

Question 8: 2×2 Table Setup (1 point)

Correct Answer: C
Either variable can form rows
Explanation: 2×2 tables are symmetric; either variable can be rows or columns.

Question 9: Fisher's Exact Test (1 point)

Correct Answer: B
When expected cell frequencies are small (less than 5 in one or more cells)

Week 9: Correlation / Regression (7 questions, 7 points)

Question 1: Correlation Coefficient Range (1 point)

Correct Answer: C
-1 to 1

Question 2: Interpreting Correlation (1 point)

Correct Answer: D
A strong negative linear relationship
Explanation: $|r| = 0.85$ is strong; negative sign indicates negative relationship.

Question 3: Correlation vs. Causation (1 point)

Correct Answer: B
The variables tend to increase together, but this does not prove causation

Question 4: Simple Linear Regression Equation (1 point)

Correct Answer: B
The slope of the regression line

Question 5: Coefficient of Determination (1 point)

Correct Answer: A
0.36
Calculation: $R^2 = r^2 = (0.60)^2 = 0.36$

Question 6: Interpreting R^2 (1 point)

Correct Answer: A
64% of the variance in y is explained by x

Question 7: Regression Assumptions (1 point)

Correct Answer: D
The independent variable (x) must be normally distributed
Explanation: This is NOT an assumption. The residuals must be normal, but x does not.

Grading Summary

Week	**MC Points**
Week 7	5 points
Week 8	9 points
Week 9	7 points
Total	**21 points**

Note: Each week also includes a group project check-in component:

- Week 7: Team Plan (5 points) - completion-based
- Week 8: Progress Update (1 point) - completion-based
- Week 9: Presentation Readiness (3 points) - completion-based